

PRACTICE EXAM

Difficulty: MEDIUM

Questions: 10

Electronics Exam

Multiple Choice Questions (4 points each)

Instructions: Choose the best answer for each question.

Question 1: According to the introduction, what is one of the concepts this lab is meant to reintroduce?

- A) Ohm's Law
- B) Kirchhoff's Laws
- C) Thevenin's Theorem
- D) Superposition Principle

Question 2: In the circuit under test, what components are connected in parallel on the right side of the circuit (before the capacitor is introduced)?

- A) Two resistors and a signal generator
- B) Two resistors and a voltage source (VCC)
- C) A resistor and a capacitor
- D) A capacitor and a voltage source

Question 3: What was the peak-to-peak voltage (vs) supplied by the signal generator in the experiments?

- A) 2 V
- B) 4 V
- C) 5 V
- D) 10 V

Question 4: According to the text, what might have caused a discrepancy in the VI waveforms between the experimental and simulated results?

- A) An incorrect capacitor value
- B) Fluctuations in the power supply
- C) The tolerance of the resistor values.
- D) Incorrect wiring of the circuit

Short Answer Questions (6 points each)

Instructions: Answer each question in 2-3 complete sentences.

Question 5: Briefly describe the purpose of measuring the left and right subcircuits separately in the experimental procedure.

Question 6: Explain in your own words how Figure 3 differs from Figure 4 in terms of the circuit configuration.

Question 7: According to Conclusion C2, what does the superposition principle state about linear circuits with multiple independent sources?

Problem-Solving Questions (10 points each)

Instructions: Show your work and explain your reasoning.

Question 8: Based on the provided tables, calculate the approximate percent difference between the RMS value of v_O in Table E1 and Table E2. Show your formula.

Question 9: Suppose the resistor in the left subcircuit was actually 550 ohms instead of 560 ohms due to component tolerances. Qualitatively, how would this difference likely affect the measured v_I (rms) value compared to the value calculated assuming a 560-ohm resistor, assuming all other factors remain constant? Explain your reasoning.

Question 10: Explain how the "negligence of the excess resistance" could contribute to the differences in the amplitudes of the waveforms of v_I , mentioned in part C3. How would you account for the extra resistance?